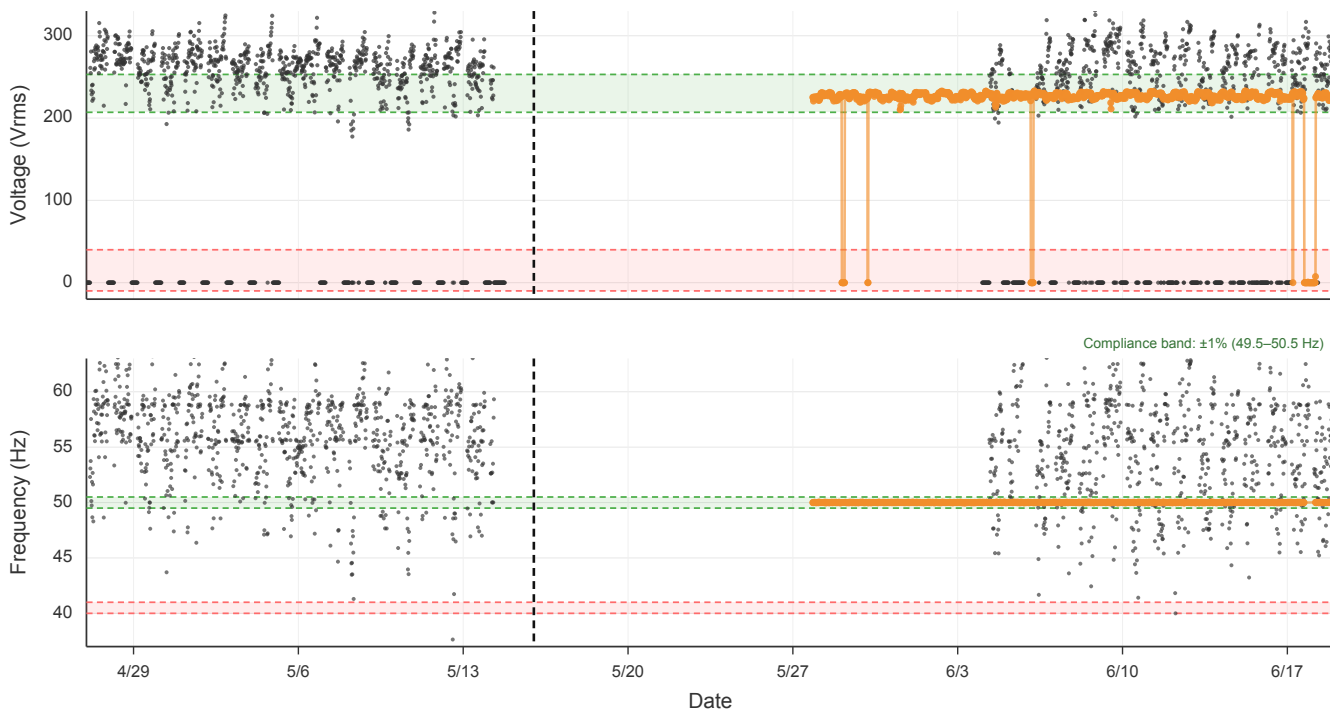


Pre-intervention:
micro-hydro grid only

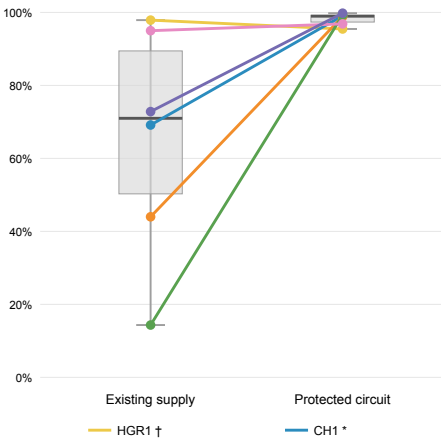
Intervention date
(May 16, 2024)

Post-intervention: stacked system
Protected PV circuit in use for critical loads;
micro-hydro remains available for other uses

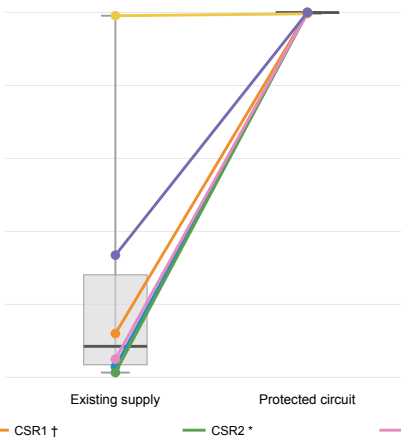
Monitoring of existing micro-hydro system



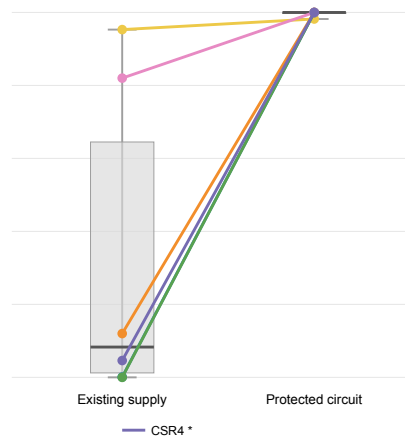
Power uptime (%)



Voltage compliance ($\pm 10\%$, 230 V)



Frequency compliance ($\pm 5\%$, 50 Hz)



* Reconstructed single-PowerWatch source comparisons; † inherited baseline values awaiting reconstruction.

Follow-up uptime: paired HOP/PowerWatch, expected-window denominator. PQR: jointly valid voltage/frequency readings.